ML4P Problem Set 1

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February 25, 2026

Github link:https://github.com/VM2708/ML4P/tree/main/Desktop/ML4P/ps1

1 Creating 3 ML models for the given spectral data

To check that my models were working, I created plots of my validation data to see if outputs of my models matched the labels. I am solely looking at the LOGG values for the data using training, test, validation sets as defined by the code. Figures 1 - 3 show the accuracy of these models.

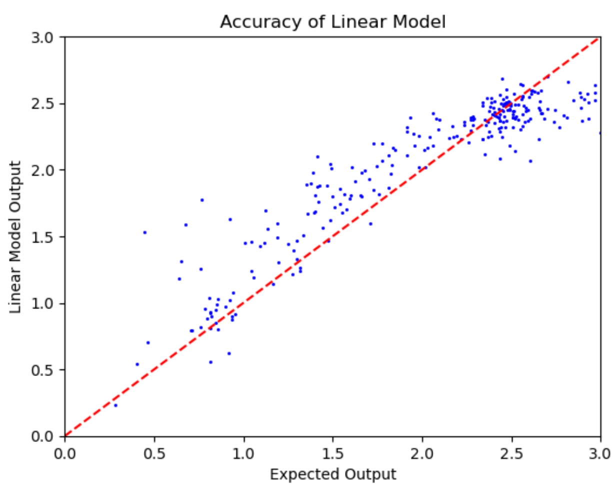

Figure 1: The blue dots represent the calculated and actual LOGG value of each validation data point. For a perfectly working model, the points should fall on the dashed red line.

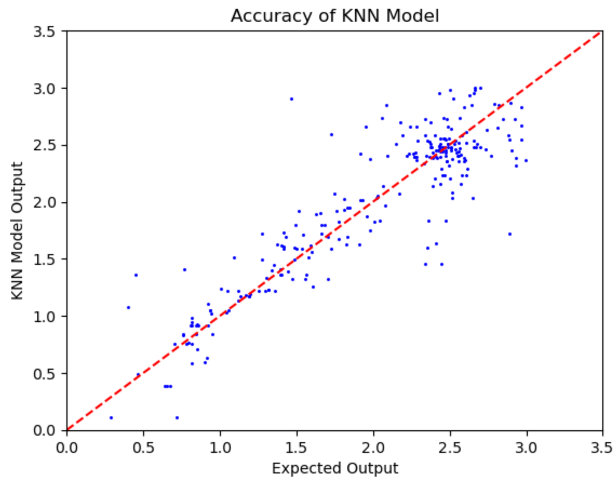

Figure 2: The blue dots represent the calculated and actual LOGG value of each validation data point. For a perfectly working model, the points should fall on the dashed red line.

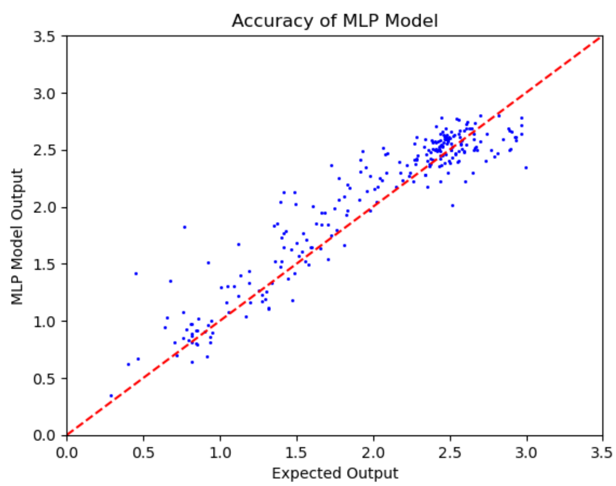

Figure 3: The blue dots represent the calculated and actual LOGG value of each validation data point. For a perfectly working model, the points should fall on the dashed red line.

2 Tuning Hyperparameters

2.1 Linear Model

Learning Rate	Batch Size	Epochs	Loss function	Optimizer fn	rnd SEED	RMSE
$1e^{-3}$	32	32	L1	Adam	42	0.8255122
$1e^{-3}$	32	32	MSE	Adam	42	0.30703128
$1e^{-4}$	32	64	MSE	Adam	42	0.3420554
$1e^{-3}$	32	128	MSE	Adam	42	0.33839915
$1e^{-3}$	32	64	MSE	Adam	42	0.3318325
$1e^{-3}$	32	32	MSE	Adam	17	0.30703128
$1e^{-3}$	32	32	MSE	Adam	6	0.287486868

Table 1: Linear Model Tuning

My final choice of hyperparameters for the linear model were the last row of table 1. This collection of hyperparameters was able to reduce my root mean squared error the most. The random seed also matters for the linear model, probably because randomization is involved in intializing the weights.

2.2 KNN

k	rnd SEED	Distance Definition	RMSE
1	17	L2	0.295859096
1	17	L1	0.20853078
10	17	L1	0.1974850
5	17	L1	0.1947782
5	5	L1	0.1947782
5	17	L1	0.1947782
5	72	L1	0.1947782

Table 2: KNN Model Tuning

My final choice of hyperparameters for the linear model were the last row of table 2. This collection of hyperparameters was able to reduce my root mean squared error the most. Changing the random seed for the KNN model does not change the final estimate as this method does not require random selection of any sort.

2.3 MLP

HiddenLayers	Activation	Rnd SEED	Loss Fnc	RMSE
[64, 32]	GELU	MSE	42	0.32682928
[64, 32]	ReLU	MSE	42	0.3981614
[64, 32]	LeakyReLU	MSE	42	0.37803027
[64, 32]	Sigmoid	MSE	42	0.6778747
[64, 32]	Tanh	MSE	42	0.68274575
[64, 32]	GELU	L1	42	0.70356745
[64, 16]	GELU	MSE	42	0.24454421
[64, 32, 16]	GELU	MSE	42	0.25378534
[64, 32, 16]	GELU	MSE	17	0.23598064
[64, 32, 16]	GELU	MSE	13	0.3519081

Table 3: MLP Model Hyperparameter tuning: Other options include, batch size, epochs, learning rate, optimizer, changing the order of activation/linear weights etc. I decided to keep those parameters the same for this testing.

My final choice of hyperparameters for the linear model were the last row of table 3. This collection of hyperparameters was able to reduce my root mean squared error the most. The random seed also matters for the MLP Model. Different seeds changed the convergence. I also chose the batch size to be 32, with 30 Epochs, and a Learning rate of $1e^{-3}$ using the Adam optimizer.

2.4 Compare Random Seed Convergence

Changing the random seed changed the convergence of the linear model and the MLP, but not the KNN model.

3 Final Test

For my final model, I decided to use K-Nearest Neighbors with 5 neighbors and the manhattan/L1 distance. These hyperparameters resulted in the lowest RMSE for the validation data set after the training. Using these hyper parameters, I got the test data results shown in figure 4 with a root mean squared error of 0.274444.

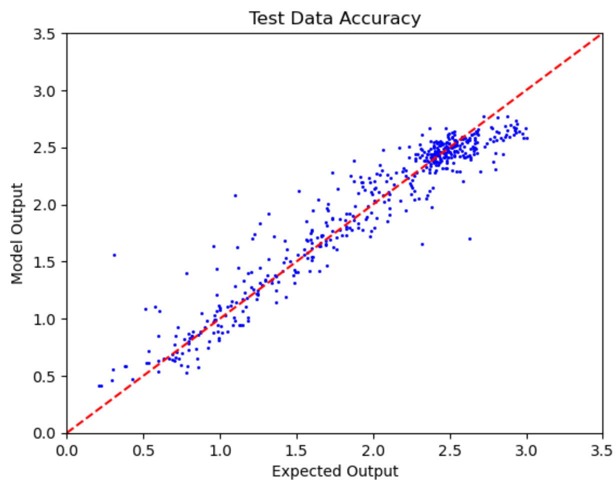

Figure 4: The Final Test shows decent correlation between the calculated output and the given LOGG values of the test data

4 Final Project Ideas

My research project requires the identification of patterns in images of a cell. I've included an example point of data as figure 5. Fluorescence corresponds to a specific protein associated with speckles, and the arrows show motion over time of this protein. Currently, I have 16 streams, each with 100 images of one cell taken in series. I currently have two ideas for ML projects with this data. My first idea would be to use ML to identify the speckles and mask out the nucleoli. The speckles are the bright spots in this image, and the nucleoli are those darker circular spots. My second idea would be to use ML to extend my data past the 100 frames so I could see if there are any patterns in the data.

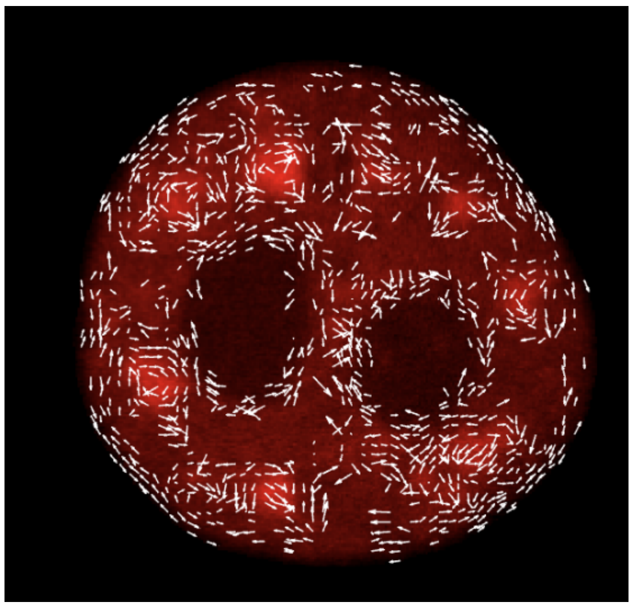

Figure 5: Example data for my research project